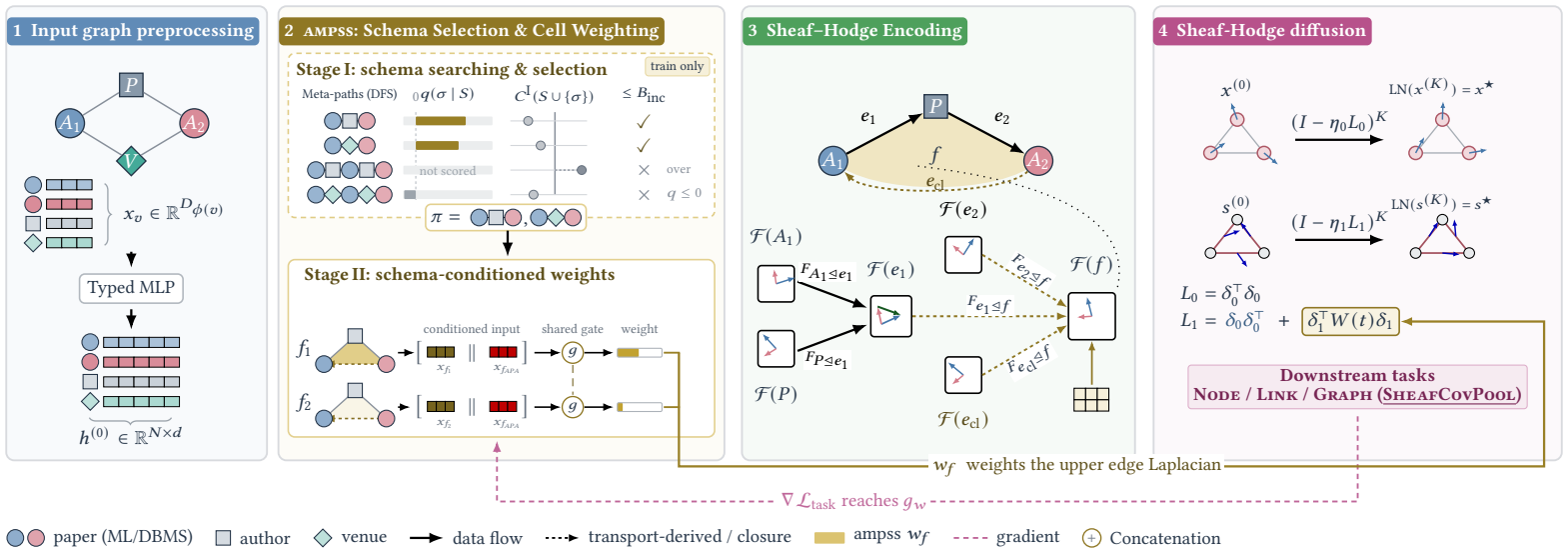

1 Input graph preprocessing
P
A1
A2
V
$x_v \in \mathbb{R}^{D_{\phi(v)}}$
Typed MLP
$h^{(0)} \in \mathbb{R}^{N \times d}$
2 AMPSS: Schema Selection & Cell Weighting
Stage I: schema searching & selection
train only
Meta-paths (DFS)
$_0q(\sigma \mid S)$
$C^{\mathrm{I}}(S \cup \{\sigma\})$
$\leq B_{\mathrm{inc}}$
not scored
over
$q \leq 0$
$\pi =$
Stage II: schema-conditioned weights
conditioned input
shared gate
weight
f_1
f_2
x_{f_1}
x_{f_2}
$x_{f_{\mathrm{APA}}}$
g
3 Sheaf–Hodge Encoding
e_1
e_2
f
e_{cl}
$\mathcal{F}(e_2)$
$\mathcal{F}(A_1)$
$\mathcal{F}(e_1)$
$\mathcal{F}(f)$
$\mathcal{F}(P)$
$\mathcal{F}(e_{\mathrm{cl}})$
$F_{A_1 \trianglelefteq e_1}$
$F_{P \trianglelefteq e_1}$
$F_{e_1 \trianglelefteq f}$
$F_{e_2 \trianglelefteq f}$
$F_{e_{\mathrm{cl}} \trianglelefteq f}$
4 Sheaf-Hodge diffusion
$x^{(0)}$
$(I - \eta_0 L_0)^K$
$\mathrm{LN}(x^{(K)}) = x^\star$
$s^{(0)}$
$(I - \eta_1 L_1)^K$
$\mathrm{LN}(s^{(K)}) = s^\star$
$L_0 = \delta_0^\top \delta_0$
$L_1 = \delta_0 \delta_0^\top + \delta_1^\top W(t) \delta_1$
Downstream tasks
NODE / LINK / GRAPH (SHEAFCOVPOOL)
w_f weights the upper edge Laplacian
$\nabla \mathcal{L}_{\mathrm{task}}$ reaches g_w
paper (ML/DBMS)
author
venue
data flow
transport-derived / closure
ampss w_f
gradient
Concatenation